Cheshire YMCA robotics program

Fall 2026 An eight session proposal for grades 5 and 6

MechaRAMS can use the YMCA's two existing LEGO Education SPIKE Prime kits to offer a small, welcoming robotics program led by trained high school mentors. Students will work in teams, test a robot every week, and learn to explain a few useful programming ideas through physical results.

Recommended pilot

12 children in grades 5–6 form four teams of three: T1/T2 share Kit A and T3/T4 share Kit B. Four student instructors cover two robot stations and two build tables; a fifth or sixth can float. The adult lead oversees the room and YMCA procedures. Bring four available laptops: two remain connected to robots and two show build references. Printed build cards make the build activity work with two laptops too.

What students should gain

By the final session, students should be able to trace an ordered program, change a variable, explain a counted loop and a sensor decision, improve a design using test results, and describe a small robot mission. Eight contact hours provide an introduction to Python and engineering practice. Independent Python fluency is beyond this pilot's scope.

Why this fits the partnership

The YMCA receives a community learning activity using its existing equipment. Younger students meet approachable high school role models. MechaRAMS develops student teaching skills and offers a voluntary path toward future high school robotics participation. Community benefit and learning are the primary outcomes; future recruitment is a longer-term outcome to track separately.

Known facts and planning assumptions

The user confirms 12 children in grades 5–6, two SPIKE kits, 4–6 student instructors and four available laptops. Student instructors practiced the materials on October 25 and requested more building, clearer roles and shorter code discussion. Dates, local procedures, inventory and hardware acceptance still need confirmation. No child learning outcomes are claimed.

A decision before announcing registration

Confirm eight dates, adult supervision, room layout, storage, the printing plan and both kits' working condition. Use the confirmed 12-child grouping and prepare four equal build packs. Inventory available parts before buying supplies.

Eight sessions with a visible result

Wk	Project	Programming focus	Evidence
1	Meet your robot	Sequence and a function call	Letter and action order
2	Delivery to a target	Variables and a fair test	Travel in cm
3	Robot dance studio	Counted loops and indentation	Movement count
4	Parking assistant	Sensor input and if conditions	Gap in mm and stop reason
5	The floor tells the robot	Calibration and repeated decisions	Reflection and detection outcome
6	A rescue attachment	Engineering design and reusable functions	Controlled deliveries out of attempts
7	Community robot mission	Decomposition and integration	Team-defined success criterion
8	Robot showcase	Explanation, evidence, and reflection	Observed result and individual explanation

A common robot with changing missions

Sessions 1 to 3 establish movement and programming habits. Session 4 adds distance sensing. Session 5 adds floor marker detection. Session 6 emphasizes a student-built attachment. Sessions 7 and 8 combine familiar ideas into a chosen mission and explanation. This progression protects testing time and avoids a new full model build every hour.

A choice for students who already know robotics

Offer additional calibration trials, a reusable function, or a more demanding success criterion. Keep the same shared challenge so students can help one another. Ask experienced students to explain their reasoning and leave the keyboard available to their partners.

A useful scope boundary

The core course uses a basic set, passive attachment, and one sensor decision at a time. A line follower, powered claw, precise gyro navigation, computer vision, and competitive FLL field missions are possible future topics, but would add setup and debugging demands to this short pilot.

Enrollment and staffing

Count	Grouping	Kit access	Recommendation
12	Four teams of three	Two teams share each robot	Confirmed class plan
4–6	Two robot and two build leads	Additional mentors float	Plus an adult lead

Staffing plan

Assign one adult program lead who follows YMCA supervision and pickup procedures. Use the confirmed 4–6 student instructors: two lead robot stations and two lead build tables. Additional mentors help with transitions and individual support. Identify a backup before class.

Device plan

Bring the four available laptops. Leave two at the robot stations and use two for build instructions or reference photos; those two do not connect to the hubs. Printed build cards support the same hands-on station with only two laptops. Test the actual devices, Python, connection and save/export workflow before class [2].

Participation and access

No prior coding experience or home robot is needed. Use floor or seated table demonstrations according to access needs, large printed code, spoken and written instructions, and flexible ways to answer. Ask families about accommodations through the normal YMCA process. Mentors rotate control so confident students cannot monopolize the robot.

Time beyond the weekly hour

Budget about two hours for the mentor workshop, plus instructor kit preparation and a complete rehearsal before launch. Each week allow roughly 20–30 minutes before the session and 15 minutes afterward for setup, charging, file backup, and debrief. These are planning estimates, not confirmed volunteer commitments.

Calendar and materials

Session dates remain unconfirmed. The earlier October 21 proposal start has passed. Agree eight actual dates and a make-up plan with the YMCA before distributing a calendar.

Week	Confirmed date	Project
1	To be agreed	Meet your robot
2	To be agreed	Delivery to a target
3	To be agreed	Robot dance studio
4	To be agreed	Parking assistant
5	To be agreed	The floor tells the robot
6	To be agreed	A rescue attachment
7	To be agreed	Community robot mission
8	To be agreed	Robot showcase

Check school holidays, Thanksgiving week, room access and the final session date when agreeing the calendar. Preserve eight one-hour sessions; do not assume the original example dates are booked.

Item	Amount for 12 students	Source or planning allowance
SPIKE Prime kits	2 complete working sets	Existing YMCA equipment; inspect first
Laptops	4 available; bring all 4	2 robot control and 2 build references
Charging and data cables	2 compatible sets	Inventory first; use data-capable USB for setup
Paper, printing, pencils, tape	Weekly team sheets, build cards, 14 pencils	Use per-session checklists; inventory owned supplies first
Floor surface, rulers, cardboard, cubes	2 lanes, 4 rulers, up to 6 cubes	Reusable kit plus weekly build supplies
Storage	2 kit bins and 4 small attachment trays	Reuse available containers

Consumables total: approximately $35–70 for planning only. These are allocation estimates, not researched vendor prices. Device purchase, replacement electronics, program fees, transportation, and staff costs are excluded. Confirm responsibility for expenses with the YMCA.

Programming approach and available materials

Real Python in small recipes

Students use complete supplied SPIKE App 3 projects. The mentor introduces only the short student recipe first. Helper functions give movement and sensing meaningful names. Students predict, run, investigate, modify, and eventually compose their own short mission. They see the underlying sensor condition and loop as those concepts become relevant.

What Python in blocks means here

The recipe cards are small pieces of real text code that students can arrange on paper and paste into an existing project. This is a teaching scaffold, not a claim that the SPIKE App offers drag-and-drop Python blocks or automatic conversion between Word Blocks and Python. Keep the setup and hardware library code supplied. Teach indentation, quotation marks, and await only as much as needed for the current task.

A planned fallback

Each lesson includes a parallel Word Blocks path with the same prediction and physical outcome. Switch a team or the class when typing or connection difficulties prevent learning. Return to the same concept later. Record which language students used so final learning claims are accurate.

Resource	Best use in this pilot	Boundary
Official app onboarding [2]	Mentor setup and device rehearsal	Test the exact installed version
Driving Around and build [3–4]	Common base and beginner movement ideas	Our sessions and port map differ
Official Python unit and guide [5–6]	Instructor background and future extensions	Full curriculum exceeds eight hours
Cart Control and Python help [7–9]	Sensor concepts and API checking	Legacy Python examples may not run unchanged
This package	Weekly teaching sequence and complete starter files	Requires local hardware rehearsal

LEGO ended direct SPIKE portfolio sales on June 30, 2026, but states that the app and lessons remain supported through June 30, 2031 [1]. That makes existing kits useful for this proposal. Future equipment expansion should be considered separately.

School connection and evidence of benefit

Complement the school program

Dodd's public program description includes autonomous robotics and programming [10]. It does not identify the current hardware, so this proposal does not assume the school still uses Mindstorms or now uses SPIKE. Before launch, ask the school contact which grades study robotics, what platform they use, and what concepts students may already have met.

A reason to choose the YMCA program

The distinctive offer is a small community mission designed with high school mentors, repeated opportunities to explain code, and an optional introduction to the high school team. Avoid advertising it as something unavailable in school until that has been checked.

Engineering Inspiration

The published criteria emphasize measurable community STEM education, recruitment, and ongoing effort [11]. This pilot can supply evidence of student-led teaching and changes in participants' knowledge or interest. Record what happened, who contributed, and how the YMCA benefited. Do not equate total attendance with unique students.

FIRST Impact Award

Impact places emphasis on sustained efforts and recent accomplishments within the last three years [12]. One new eight-week pilot can be part of the team's story, but does not by itself establish sustained impact. Plan a repeat cohort, preserve curriculum revisions, and ask the partner whether the program met a real need. Recheck current season submission instructions before applying. The published 2027 deadline was February 11 at 3 p.m. ET when checked on October 1, 2026.

Follow the students over time

Offer a parent-mediated invitation to a future open house. Record opt-in interest separately from attendance and later actual team participation. Children in grades 5 and 6 are several years from high school. The program should remain valuable for children who never join FRC.

Pilot success indicators

Use proposed targets as planning goals: all 12 children get an operating turn each week; at least nine of twelve attend six or more sessions; returning students can explain at least three of four core concepts; the YMCA requests a repeat or gives specific useful feedback. These are proposed goals, not achieved results. Report actual attendance, matched responses and missing data.

Sources and next decisions

Sources checked October 1, 2026. Links are clickable in this document. This curriculum's activities and code are original adaptations, not official LEGO or FIRST course materials. The supplied LEGO build PDF remains LEGO's work.

[1] LEGO SPIKE portfolio update

SPIKE sales ended June 30, 2026. App and curriculum support continues through June 30, 2031. Existing kits are suitable for this pilot.

[2] LEGO SPIKE App download

Official software entry point. Check the requirements for the exact proposed devices and test Python, connection, and project saving before enrollment.

[3] LEGO Training Camp 1 Driving Around

[4] LEGO Practice Driving Base instructions

Official illustrated assembly sequence. Follow the complete build before the pilot, then cable the left motor to A and right motor to B for our code.

[5] LEGO Python Communicating Ideas unit

[6] LEGO Python teacher guide Units 1 to 5

[7] LEGO Cart Control

[8] LEGO live Python help

Final authority for the installed SPIKE App Python API. Open Help in a Python project if the direct link does not display.

[9] Tufts CEEO SPIKE 3 documentation archive

[10] Dodd Program of Studies Career and Technical Education

Describes robotics and programming learning. The public description does not establish which robot kit is currently used. Ask the school before claiming alignment or novelty.

[11] FIRST Team Attribute Awards

Engineering Inspiration emphasizes measurable STEM recruitment and community education with ongoing effort. It is normally evaluated through team interactions rather than a separate Impact-style essay.

[12] FIRST Submitted Awards

Impact recognizes sustained work, with particular attention to the last three years. The page lists a February 11, 2027, 3 p.m. ET Impact deadline as checked October 1, 2026. Recheck current season instructions.

[13] FIRST judging best practices

The adult coordinator still needs to confirm dates, storage, charging and local procedures, and complete the two-kit rehearsal. Enrollment, grade band, available mentors and four available laptops were confirmed October 25. Use the class preparation checklists for printing and packing.

Confirmed class format

12 children in grades 5–6 form four teams of three: T1/T2 share Kit A and T3/T4 share Kit B. Four student instructors cover two robot stations and two build tables; a fifth or sixth can float. The adult lead oversees the room and YMCA procedures. Bring four available laptops: two remain connected to robots and two show build references. Printed build cards make the build activity work with two laptops too.

The course introduces robotics through playful missions, student-built attachments, testing and explanation. Small Python recipes support those activities. The two drive bases stay assembled; children build a different removable feature or course model each week. Bring owned supplies after checking the per-session preparation lists. Quantities in those lists are packing targets, not confirmed inventory.

Week	Hands on build
1	A removable robot mascot
2	A cargo tray that keeps its load
3	A removable cone pusher
4	A garage with a dependable target wall
5	A road with a delivery marker
6	A rescue scoop or guide
7	A small community mission scene
8	A working exhibit for the showcase